



(i) Trace the program currently in memory using the following trace table.

| Instruction address | ACC | Memory address |    |    |    |    |
|---------------------|-----|----------------|----|----|----|----|
|                     |     | 10             | 11 | 12 | 13 | 14 |
|                     |     | 12             | 11 | 10 | 22 | 22 |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |
|                     |     |                |    |    |    |    |

[3]

(ii) State the effect of changing instruction `LDD 10` in address 100 to `LDM #10`

.....  
 .....  
 ..... [1]

(iii) Identify **and** describe **one** mode of addressing **not** given in the table of instructions in part (a).

Mode of addressing .....

Description .....

.....  
 ..... [2]

